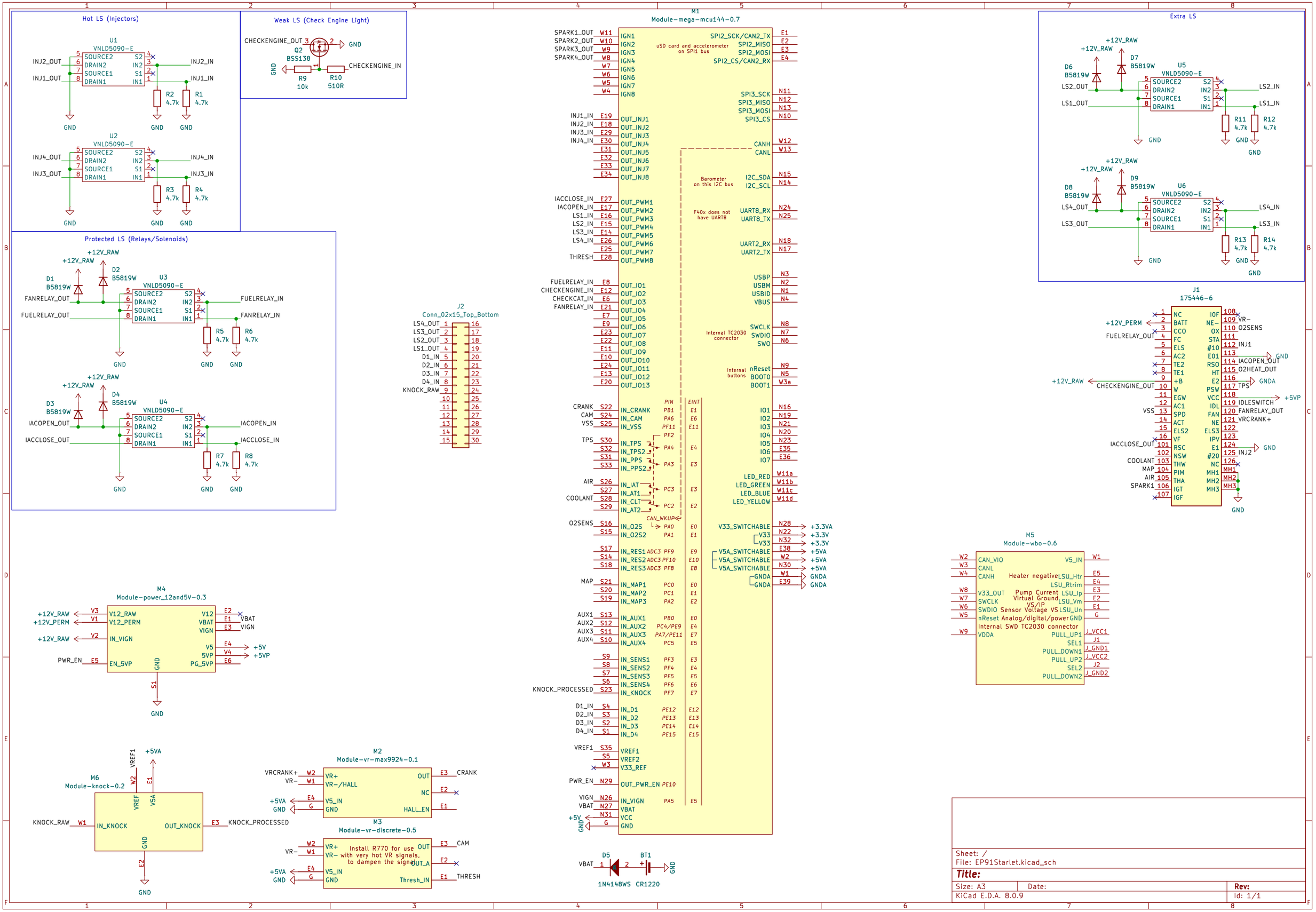
Hot LS (Injectors)
U1
VNLD5090-E
INJ2_OUT
INJ1_OUT
INJ2_IN
INJ1_IN
R2 4.7k
R1 4.7k
U2
VNLD5090-E
INJ4_OUT
INJ3_OUT
INJ4_IN
INJ3_IN
R3 4.7k
R4 4.7k
Weak LS (Check Engine Light)
CHECKENGINE_OUT
Q2
BSS138
GND
R9 10k
R10 510R
CHECKENGINE_IN
Protected LS (Relays/Solenoids)
+12V_RAW
D1 B5819W
D2 B5819W
U3
VNLD5090-E
FANRELAY_OUT
FUELRELAY_OUT
FUELRELAY_IN
FANRELAY_IN
R5 4.7k
R6 4.7k
D3 B5819W
D4 B5819W
U4
VNLD5090-E
IACOPEN_OUT
IACCLOSE_OUT
IACOPEN_IN
IACCLOSE_IN
R7 4.7k
R8 4.7k
M1
Module-mega-mcu144-0.7
SPARK1_OUT
SPARK2_OUT
SPARK3_OUT
SPARK4_OUT
uSD card and accelerometer on SPI1 bus
SPI2_SCK/CAN2_TX
SPI2_MISO
SPI2_MOSI
SPI2_CS/CAN2_RX
SPI3_SCK
SPI3_MISO
SPI3_MOSI
SPI3_CS
CANH
CANL
Barometer on this I2C bus
I2C_SDA
I2C_SCL
F40x does not have UART8
UART8_RX
UART8_TX
UART2_RX
UART2_TX
USBP
USBM
USBID
VBUS
Internal TC2030 connector
SWCLK
SWDIO
SWO
Internal buttons
nReset
BOOT0
BOOT1
LED_RED
LED_GREEN
LED_BLUE
LED_YELLOW
V33_SWITCHABLE
V5A_SWITCHABLE
GNDA
CAN_WKUP
M2
Module-vr-max9924-0.1
VRCRANK+
VR-
VR+
VR-/HALL
V5_IN
GND
OUT
NC
HALL_EN
CRANK
M3
Module-vr-discrete-0.5
Install R770 for use with very hot VR signals, to dampen the signal
OUT_A
Thresh_IN
CAM
THRESH
M4
Module-power_12and5V-0.3
V12_RAW
V12_PERM
IN_VIGN
EN_5VP
V12
VBAT
VIGN
V5
5VP
PG_5VP
PWR_EN
M5
Module-wbo-0.6
CAN_VIO
CANL
CANH
V5_IN
Heater negative
LSU_Htr
LSU_Rtrim
V33_OUT
Pump Current
Virtual Ground
LSU_Ip
LSU_Vm
SWCLK
SWDIO
Sensor Voltage
LSU_Un
nReset
Analog/digital/power
GND
Internal SWD TC2030 connector
VDDA
PULL_UP1
SEL1
PULL_DOWN1
PULL_UP2
SEL2
PULL_DOWN2
M6
Module-knock-0.2
VREF
V5A
IN_KNOCK
OUT_KNOCK
KNOCK_RAW
KNOCK_PROCESSED
J2
Conn_02x15_Top_Bottom
LS4_OUT
LS3_OUT
LS2_OUT
LS1_OUT
D1_IN
D2_IN
D3_IN
D4_IN
KNOCK_RAW
Extra LS
D6 B5819W
D7 B5819W
U5
VNLD5090-E
LS2_OUT
LS1_OUT
LS2_IN
LS1_IN
R11 4.7k
R12 4.7k
D8 B5819W
D9 B5819W
U6
VNLD5090-E
LS4_OUT
LS3_OUT
LS4_IN
LS3_IN
R13 4.7k
R14 4.7k
J1
175446-6
+12V_PERM
FUELRELAY_OUT
CHECKENGINE_OUT
IACCLOSE_OUT
COOLANT
MAP
AIR
SPARK1
VR-
O2SENS
O2HEAT_OUT
TPS
IDLESWITCH
FANRELAY_OUT
VRCRANK+
INJ1
IACOPEN_OUT
INJ2
GNDA
+5VP
D5
BT1
1N4148WS
CR1220
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